

Kushal Jain

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PROFESSIONAL SUMMARY

Engineering-focused experimental physicist delivering Pound-Drever-Hall (PDH) locking, system commissioning, noise diagnostics, and precision measurement workflows. Contributed to the commissioning of power-recycled Michelson interferometers achieving 1.5 kW circulating optical power, $\sim 370\times$ power build-up, and order-of-magnitude vibration noise suppression. Experienced in optical system design, ultra-high vacuum operation, hardware-integrated calibration, and Python-based signal analysis. Caltech LIGO Laboratory research fellow with experience in precision instrumentation.

EDUCATION

PhD in Physics

Jan 2025 – Present

Cardiff University, UK | STFC-funded

Supervisors: Prof H. Grote · Prof K. Dooley · Dr K. Kokeyama

Post-Graduate Diploma, Physics & Astronomy

2023 – 2024

Ashoka University, India | CGPA 3.62/4.0 · Cum Laude

B.Sc (Honours) Physics

2020 – 2023

Ashoka University, India | CGPA 3.36/4.0 · Upper Second Class equivalent

RESEARCH EXPERIENCE

PhD Researcher

Jan 2025 – Present

Gravity Exploration Institute, Cardiff University

- **Interferometer Commissioning:** Commissioned power-recycled Michelson interferometers for high-sensitivity measurements. Implemented laser frequency stabilization using PDH locking (dither and DC-readout loops), achieving 1.5 kW circulating optical power and $\sim 370\times$ power build-up through optimized lock acquisition.
- **Vibration Isolation & Diagnostics:** Diagnosed mechanical coupling pathways using DARM transfer-function measurements. Led the implementation of passive elastomeric vibration isolation using Viton damping pads, achieving a $\sim 10\times$ RMS noise reduction above 45 Hz and enabling a $10\times$ control loop gain increase.
- **Control Systems & Vacuum Operation:** Measured open-loop transfer functions and designed digital suppression notch filters to improve system stability. Operated ultra-high vacuum (UHV) systems to 10^{-7} mbar and maintained structured commissioning logs for reproducible operation.
- **Optical Calibration Hardware:** Designed an independent optical calibration system via phase-modulated scattered-light injection. Engineered motor mounts and cog-wheel hardware in OpenSCAD for PLA+ 3D printing, integrating Python and Arduino Uno for drive/acquisition control in the MHz regime.
- **Birefringence Fluctuation Measurements in Crystalline Silicon:** Redesigned experimental approach for improved noise characterization and performed sensitivity modeling and noise budget analysis to evaluate compliance with next-generation gravitational-wave detector requirements.

Summer Research Fellow — LIGO Laboratory, California Institute of Technology

May – July 2023

NSF/Caltech-funded Program | Mentors: Dr Y. Michimura, Prof R. Adhikari

- Designed and executed high-sensitivity polarimetry experiment to measure birefringence fluctuations in crystalline silicon, validating shot-noise-limited behavior in the measurement band.
- Established first upper-limit measurements of birefringence fluctuations in silicon and assessed compliance with next-generation detector performance specifications.

Undergraduate Thesis Researcher

Sept 2023 – May 2024

Ashoka University | Mentor: Prof Somak Raychaudhary

- Analyzed large astrophysical datasets to investigate correlations between galaxy properties across redshift bins

TECHNICAL SKILLS

Hardware & Optics	Laser frequency stabilization using PDH and Fabry-Pérot cavity locking; feedback control system tuning; vibration isolation; high-precision optical alignment; noise budget analysis; notch filter design; ultra-high vacuum (UHV) operation; system integration; cleanroom protocols
Software & Analysis	Python (NumPy, SciPy, Matplotlib); spectral analysis (ASD/PSD); Git; Linux; CAD (OpenSCAD)
Instrumentation & Control	Moku:Pro (signal analyzer, PID, lock-in); control & data system; electro-optic components (EOM/AOM); PZT and piezomotor actuators; high-bandwidth photodetectors; GigE cameras

AWARDS & FUNDING

• STFC PhD Studentship	2025–2029
Fully funded UKRI doctoral scholarship in gravitational-wave physics (Science & Technology Facilities Council)	
• Visiting Researcher — LIGO Livingston Observatory, Louisiana, USA	July–Sept 2026
Funded research placement at large-scale interferometric gravitational-wave detector facility	
• STFC Visiting Scholar	2026–27
Competitive funded international research placement	
• NSF LIGO SURF Fellowship — Caltech LIGO Laboratory, USA	2023
NSF-funded undergraduate research fellowship	
• Cum Laude	2024
Graduated with distinction, Post-Graduate Diploma in Physics & Astronomy	

PRESENTATIONS & PUBLICATIONS

• “Optical Calibration of Gravitational-Wave Detectors using Backscattered Light”	Sept 2024
UK Gravitational-Wave Conference, University of Nottingham <i>Contributed Talk</i>	
• “Optical Calibration Systems”	Jan 2026
Nordic Winter School on Black Holes & Gravitational Waves, Norway <i>Presentation Talk</i>	
• “Measuring Birefringence and Its Fluctuations in Crystalline Silicon”	July 2023
Research Presentation, Caltech LIGO Laboratory (30 scientists)	

TEACHING EXPERIENCE

- **Teaching Assistant, Cardiff University** (2025–2026): Optics and Electromagnetism courses; assignment marking, course material preparation, and laboratory support.
- **Teaching Assistant, Ashoka University** (2023–2024): Astronomy Laboratory and Principles of Science courses; delivered tutorials and supported laboratory instruction for undergraduate cohorts.

LEADERSHIP & SERVICE

- **Founding Secretary**, The Astronomy Society, Ashoka University (Apr 2023–Jun 2024): Organized India’s first inter-university Radio Astronomy Conference (70+ participants).
- **Student Mentor**, Ashoka University (Aug 2022–Jun 2024): Mentored undergraduate students on academic planning and graduate study preparation.